6-Member Team Video Presentation Guide (Strict 6-Section Rubric)

Course: Explainable AI (DA351) — Phase 2 Evaluation | **Max Duration:** Exactly 10:00 Minutes
Paper: *Case-Based Explanation with Feature Highlighting* (Kenny et al., IJCAI 2023) | **Dataset:** CUB-200-2011

1. Evaluation Mapping: 6 Evaluation Sections Mapped to 6 Team Members

Speaker	Mandatory Evaluation Section	Slides	Time	Timestamp
Member 1	**1. Introduction:** Paper title, authors, problem addressed & research objective	Slides 1 & 2	**1 min**	0:00 – 1:00
Member 2	**2. Research Gap:** Published results audit (pp. 5-6), 7x7 resolution & metric crash	Slides 3 & 4	**2 min**	1:00 – 3:00
Member 3	**3. Proposed Improvement:** Multi-Scale LayerCAM, Cosine matching & architecture flow	Slides 5 & 6	**2 min**	3:00 – 5:00
Member 4	**4. Implementation:** PyTorch gradient hooks, logit handling & 2x2 factorial ablation setup	Slide 7	**2 min**	5:00 – 7:00
Member 5	**5. Experimental Results:** Bird #3 visual proof, faithfulness AUC, metric ablation & table	Slides 8–11	**2 min**	7:00 – 9:00
Member 6	**6. Conclusion:** Key findings, technical contributions, acknowledged limitations & Q&A;	Slide 12	**1 min**	9:00 – 10:00

2. Video Recording Checklist & Team Coordination Guide

Step	What to Do (Simple Instructions)
1. Screen Setup	Open Phase_2_Project_Presentation.pptx in Slide Show mode (or open presentation_slides.html in Chrome/Edge and press F11).
2. Transitions	Always say your handoff sentence at the end of your section (e.g., *"I now hand over to Member 3"*). This proves strong team synergy.
3. Timekeeping	Keep a phone stopwatch on silent. Members 1 and 6 speak for **1 minute**; Members 2, 3, 4, and 5 speak for **2 minutes**. Total time = 10:00 minutes.
4. Audio Check	If recording over Zoom or Google Meet, test all 6 microphones before starting the final recording to ensure balanced audio levels.
5. Drive Sharing	Upload the recorded MP4 file to Google Drive. Right-click, select **Share**, and change permissions to **"Anyone with the link can view"**.

3. Complete Submission Inventory (Inside submission_phase2/)

Deliverable Folder	Key Files Inside	Evaluation Purpose
A_Video_Presentation_Assets/	• Phase_2_Project_Presentation.pptx • 6_Member_Team_Presentation_Guide.pdf • presentation_slides.html	12-slide master deck with visual point-to cues, spoken scripts, motives, and recording guidelines.
B_Original_Research_Article/	• Original_Research_Article_IJCAI2023_Kenny_et_al.pdf	Complete PDF of original paper (Kenny et al., IJCAI 2023, Pages 3142–3150) as required.
C_Experimental_Results_and_Comparison/	• Experimental_Results_and_Comparison.pdf	8-page comparison report with published extracts (pp. 1, 2, 5, 6), before/after tables, and literature.
Updated_Source_Code/	• reproduce_results.py • README.md & MODIFIED_FILES.md	Executable source code implementing LayerCAM and Cosine Matching with one-click reproduction.

MEMBER 1: SECTION 1: INTRODUCTION | Rubric Time: 1 Min (0:00 – 1:00) | Slides: Slides 1 & 2

Visual Cues (What to Point to on Screen): Slide 1 (Target Paper and Objectives card) and Slide 2 (Figure 1 extract from Page 2 of Kenny et al.).

What to Say to Sir (Plain Conversational English):
"Good morning, Sir and peers. Today, our 6-member team is presenting our Phase 2 project in Explainable AI (DA351).
*Our work is an audit, technical limitation analysis, and empirical extension of the IJCAI 2023 paper titled **'Advancing Post-Hoc Case-Based Explanation with Feature Highlighting'**, authored by **Eoin Kenny, Eoin Delaney, and Mark Keane**.*
***[Slide 1]** Deep neural networks are black boxes. Presenting only a confidence percentage or a raw image does not tell the user what specific feature justified the prediction.*
***[Slide 2]** To solve this, Kenny et al. introduced a **Twin-Explanation framework**: when classifying an image, the system retrieves a similar past training exemplar and draws bounding boxes on both images to highlight the shared anatomical trait, like a sharp yellow beak.*
They implemented this using standard CAM on terminal layer4 of ResNet-34 and Euclidean distance. While the concept is intuitive, our technical audit revealed two foundational flaws in their design."
■ ***The Motive:** Establish the paper details, problem context, and twin-explanation mechanism clearly in the first minute.*
***Handoff to next person:** "I will now hand over to Member 2, who will audit the published results and explain the two technical research gaps."*

If Sir asks: "What is a Twin Explanation, and why is it better than standard Grad-CAM heatmaps?"
You answer: "Sir, standard heatmaps only show colored blobs on the test image, leaving the user guessing. A Twin Explanation pairs the test bird with a concrete historical training exemplar that shares the exact same body part, providing both visual localization and analogical proof."

MEMBER 2: SECTION 2: RESEARCH GAP IDENTIFICATION | Rubric Time: 2 Min (1:00 – 3:00) | Slides: Slides 3 & 4

Visual Cues (What to Point to on Screen): Slide 3 (Screenshots of Page 5 Figure 2 & Page 6 Figure 3 from paper) and Slide 4 (Gap 1 and Gap 2 diagnostic cards).

What to Say to Sir (Plain Conversational English):
"Thank you, Member 1. Sir, in accordance with the mandatory submission guidelines, we first audited the original paper's published experimental results:
***[Slide 3]** On **Page 5, Figure 2**, Kenny et al. evaluated occlusion and inclusion across 500 CUB images, concluding that CAM outperformed random baselines. On **Page 6, Figure 3**, they published user evaluation results showing feature highlighting improved human decision accuracy.*
[Slide 4: The Two Research Gaps]
***1. Research Gap 1 (Coarse 7x7 Grid - The Thick Highlighter):** Kenny et al. extracted CAM exclusively from layer4. On a 224x224 image, this produces a coarse 7x7 grid (32-pixel stride) with an effective receptive field exceeding 100 pixels. On fine-grained birds, their 48x48 pixel box blurs the beak, eye, neck, and water together!*
***2. Research Gap 2 (Distance Metric Breakdown - The Broken Ruler):** To get sharper resolution (14x14), earlier layer3 features must be combined with layer4 into a 768-d hypercolumn. But Kenny et al. used unnormalized Euclidean distance. In high dimensions, deep layer4 numbers are much larger, completely drowning out sharp layer3 spatial features and causing matching accuracy to collapse from 51.8% to 37.2%!"*
■ ***The Motive:** Audit published claims with exact page numbers and justify why resolving the 7x7 grid and Euclidean crash is critical.*
***Handoff to next person:** "I now invite Member 3 to present our proposed technical solution and supporting research literature."*

If Sir asks: "Why did matching accuracy collapse to 37.2% under Euclidean distance?"
You answer: "Sir, in 768-dimensional space, layer4 vectors have much larger numerical magnitudes than layer3. Euclidean distance measures straight-line length, so layer4 overpowered layer3 and drowned out the sharp spatial features. Cosine similarity evaluates pattern angle, restoring balance."

MEMBER 3: SECTION 3: PROPOSED IMPROVEMENT | Rubric Time: 2 Min (3:00 – 5:00) | Slides: Slides 5 & 6

Visual Cues (What to Point to on Screen): Slide 5 (LayerCAM & Cosine similarity equations with literature citations) and Slide 6 (End-to-end architecture pipeline diagram).

What to Say to Sir (Plain Conversational English):
"Thank you, Member 2. Sir, here is our proposed technical solution grounded in top peer-reviewed literature:
[Slide 5: Proposed Improvement & Literature]
*• **To solve Gap 1 (Coarse Resolution):** We implemented **Multi-Scale LayerCAM** (published by Jiang et al. in IEEE Transactions on Image Processing, 2021). Standard CAM averages gradients across entire channels, washing out fine details. LayerCAM calculates individual positive weights for every pixel coordinate. We fuse intermediate layer3 (14x14) and layer4 (7x7) to quadruple grid density from 49 to 196 cells!*
*• **To solve Gap 2 (Metric Breakdown):** We introduced **Scale-Invariant Normalized Cosine Matching**, supported by Hypercolumn representations from Hariharan et al., CVPR 2015. We normalize 768-d hypercolumns onto the unit hypersphere so that similarity evaluates pattern angle rather than vector length, giving both layers equal voice.*
***[Slide 6: Architecture Pipeline]** Tracing the diagram: the input bird flows through ResNet-34. Saliency maps from layer3 and layer4 are bilinearly fused into a sharp 14x14 grid. At the peak coordinate, the 768-d hypercolumn is normalized and matched via Cosine similarity to retrieve the twin exemplar."*
■ ***The Motive:** Anchor the proposed improvements directly in established IEEE TIP and CVPR computer vision literature.*
***Handoff to next person:** "I will now pass to Member 4, who will explain how we implemented this in PyTorch and structured our controlled factorial experiment."*

If Sir asks: "How does LayerCAM differ from standard Grad-CAM?"
You answer: "Sir, Grad-CAM computes a single global average weight per feature map, which blurs out spatial details. LayerCAM assigns an individual weight to every spatial pixel coordinate using positive gradients, preserving fine-grained localized structures like bird beaks."

MEMBER 4: SECTION 4: IMPLEMENTATION | Rubric Time: 2 Min (5:00 – 7:00) | Slides: Slide 7

Visual Cues (What to Point to on Screen): Slide 7 (Left: PyTorch implementation highlights; Right: Controlled 2x2 Factorial ablation condition matrix).

What to Say to Sir (Plain Conversational English):
"Thank you, Member 3. Sir, let us examine how our improvement was technically implemented and what changes were made to the original method:
[Slide 7: Technical Implementation]
*1. **Preserving the Backbone:** We kept the pretrained ResNet-34 (SavedNetC) checkpoint 100% frozen at 83.60% top-1 accuracy on CUB-200. Our explanation method is strictly post-hoc.*
*2. **PyTorch Gradient Hooks:** We attached forward and backward hooks to both layer3 (14x14, 256-d) and layer4 (7x7, 512-d).*
*3. **Handling 3-Tuple Logits:** In the original codebase, SavedNetC returns a 3-tuple `(logits, x, C)`. We properly routed backward gradients exclusively through the target class logit.*
*4. **Zero-Leak Memory Safety:** By immediately detaching tensors and clearing hooks, peak GPU VRAM remained flat at 286.8 MB across all 500 images.*
*5. **Controlled 2x2 Factorial Design:** To scientifically isolate resolution effects from distance metric effects, we evaluated all 4 factorial combinations on the identical 500 CUB test images: Condition 1 (Baseline 7x7 + L2), Condition 2 (7x7 + Cosine), Condition 3 (14x14 LayerCAM + L2), and Condition 4 (Proposed 14x14 LayerCAM + Cosine)."*
■ ***The Motive:** Demonstrate rigorous software engineering, memory management, and clean 2x2 factorial experimental design.*
***Handoff to next person:** "I now invite Member 5 to present our experimental results, before-and-after visual proofs, and benchmark tables."*

If Sir asks: "Why did you use a 2x2 factorial ablation design?"
You answer: "Sir, because we introduced two technical changes: spatial resolution and distance metric. Testing all 4 combinations proves scientifically that both changes are necessary—showing that increasing resolution alone causes Euclidean distance to crash, and Cosine matching fixes it."

MEMBER 5: SECTION 5: EXPERIMENTAL RESULTS | Rubric Time: 2 Min (7:00 – 9:00) | Slides: Slides 8, 9, 10, 11

Visual Cues (What to Point to on Screen): Slides 8 (Bird #3 visual proof), Slide 9 (Deletion/Insertion curves), Slide 10 (Metric ablation bar chart), Slide 11 (Master Comparison Table).

What to Say to Sir (Plain Conversational English):
"Thank you, Member 4. Sir, here are our empirical experimental results evaluated across 500 CUB test images:
[Slide 8: Before vs. After on Bird #3] *Panel B shows Kenny et al.'s baseline: a massive 48x48 box that blurs the beak, eye, and water together. Panel C shows our proposed LayerCAM: a crisp 32x32 box focused precisely on the needle-like bill tip with zero background noise! Panel D confirms the retrieved training exemplar matches the exact same beak landmark.*
[Slide 9: Faithfulness Perturbation Game] *If an explanation is truthful, blurring its highlighted pixels must drop model confidence immediately. Over 22 steps of continuous Gaussian blur, our Deletion AUC dropped from 0.1171 to 0.0989 (t = -8.88, p = 1.22 x 10^-17), and Insertion AUC rose from 0.6817 to 0.6989 (p = 1.40 x 10^-14). In 79.6% of test images, our method was causally more faithful than Kenny's baseline!*
[Slide 10: Distance Metric Ablation] *In the bar chart, look at Condition 3: upgrading to 14x14 with Euclidean distance caused matching accuracy to collapse to 37.20% (-14.6 pp crash, p = 5.8 x 10^-8). Our Normalized Cosine Matching recovered accuracy to 46.60% (+9.40 pp recovery, p = 2.48 x 10^-6)!*
[Slide 11: Master Comparison & Hardware Profile] *Our side-by-side table confirms quadrupled resolution, statistically proven faithfulness, and recovered metric accuracy. On an NVIDIA RTX 3050 Laptop GPU, per-query latency was 488 ms (+1.4% overhead) with an identical 286.8 MB peak VRAM footprint!"*
■ ***The Motive:*** *Present empirical before-and-after proof, causal perturbation game, metric ablation bar chart, and hardware efficiency.*
Handoff to next person: *"I will now hand over to Member 6 to summarize our contributions, acknowledge limitations, and conclude our presentation."*

If Sir asks: "Why is a lower Deletion AUC better?"
You answer: "Sir, in the deletion game, we remove the pixels the method highlighted as most important. If those pixels really drove the model's prediction, the model's confidence should drop immediately. A lower area under the deletion curve proves superior causal faithfulness."

MEMBER 6: SECTION 6: CONCLUSION | Rubric Time: 1 Min (9:00 – 10:00) | Slides: Slide 12

Visual Cues (What to Point to on Screen): Slide 12 (Core Technical Contributions card, Acknowledged Limitations card, and Thank You / Q&A; banner).

What to Say to Sir (Plain Conversational English):
"Thank you, Member 5. Sir, let us synthesize our key findings, acknowledge limitations, and conclude our Phase 2 project:
[Slide 12: Key Findings & Contributions]
1. ***Quadrupled Spatial Resolution:*** *We upgraded attribution from 7x7 (49 cells) to 14x14 (196 cells) via Multi-Scale LayerCAM, shrinking bounding boxes from 48x48 to 32x32 pixels.*
2. ***Statistically Proven Faithfulness:*** *Highly significant gains in Deletion AUC (0.0989 vs 0.1171, p = 1.22 x 10^-17) and Insertion AUC (0.6989 vs 0.6817, p = 1.40 x 10^-14).*
3. ***Solved Metric Breakdown:*** *Scale-Invariant Normalized Cosine Matching recovered +9.40 percentage points over Euclidean distance in 768-d hypercolumn space (p = 2.48 x 10^-6).*
4. ***Production Efficiency:*** *488 ms per image on an entry-level RTX 3050 GPU with zero added memory overhead.*
5. ***All Deliverables Complete:*** *Video presentation assets, Original IJCAI 2023 Paper PDF, 8-page Comparison Report PDF, and reproducible source code.*
[Acknowledged Limitations & Future Work] *CUB ground truth landmarks are discrete 1D points rather than dense 2D masks; future work will test segmentation masks and Vision Transformers.*
Thank you very much, Sir! Our entire team is now ready for your questions and discussion."
■ ***The Motive:*** *Deliver a crisp, authoritative summary of contributions and limitations, leaving a polished professional impression.*
Handoff to next person: *"Thank you very much, Sir. Our entire team is now ready for your questions and open discussion!"*

If Sir asks: "What is the single biggest takeaway from your Phase 2 project?"
You answer: "Sir, our biggest takeaway is that post-hoc explanations should never rely on the terminal layer alone. By fusing multi-scale layers with LayerCAM and using Normalized Cosine Matching, we achieve sharp, faithful explanations with virtually no extra computational cost."